

Conjugated Systems & Aromaticity

Ingyu Bahng

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1 Aromaticity & Aromatic Substitution

1.1 Aromatic Compounds

Core Concept 1.1: Aromatic Compounds

Name originates from these compounds' nice scent, but now it is a strictly structural definition covered in Core Concept 1.5.

While not limited to the base benzene structure, the extent of these notes will mostly refer to base benzene compounds such as the following.

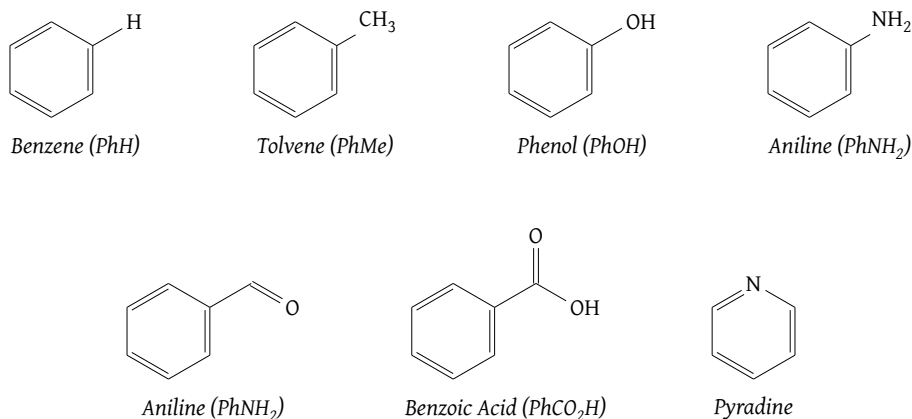


Figure 1: Phenyl ring based aromatic compound examples; pyridine represents an aromatic compound with a non CH as part of its ring

Core Concept 1.2: Aromatic Compound Stereochemistry

Additional substituent positioning to the the root aromatic compounds shown in FIG REFERENCE can be specified through the ortho, meta, and para keywords.

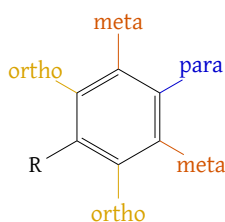


Figure 2: Phenyl ring substituent positioning keywords

The compound naming scheme follows a structure of *realtion-sub-root*. Take a look at the following examples.

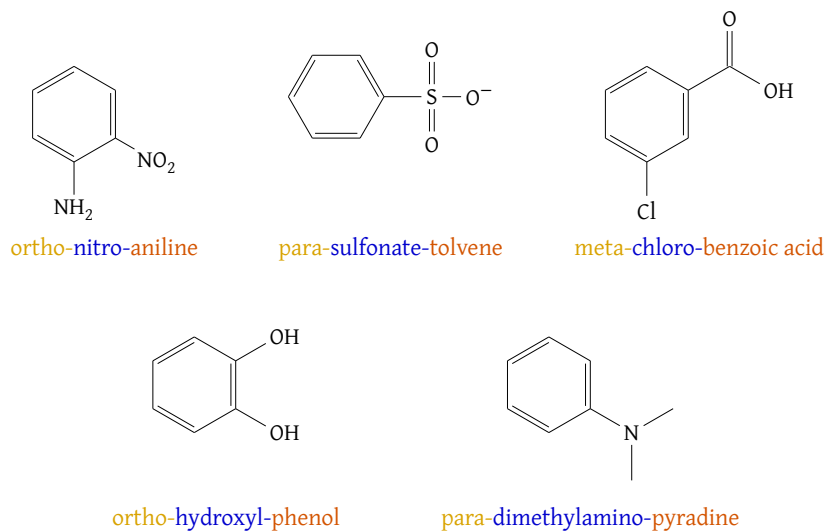


Figure 3: Phenyl ring based aromatic compound nomenclature

Core Concept 1.3: Reactivity of Aromatic Compounds

Aromatic compounds are less reactive than predicted. For instance while a simple cyclohexene reacts with mCPBA to form a secondary ester ring, no reaction occurs when benzene and 3 mCPBA are mixed together.

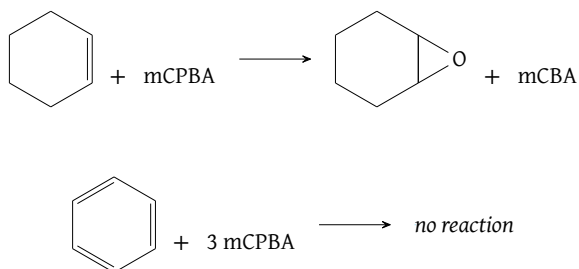


Figure 4: Cyclohexene and benzene reaction with mCPBA

Similarly, a portion of the predicted energy drop during the hydrogenation of benzene is eliminated due to aromatic stabilization.

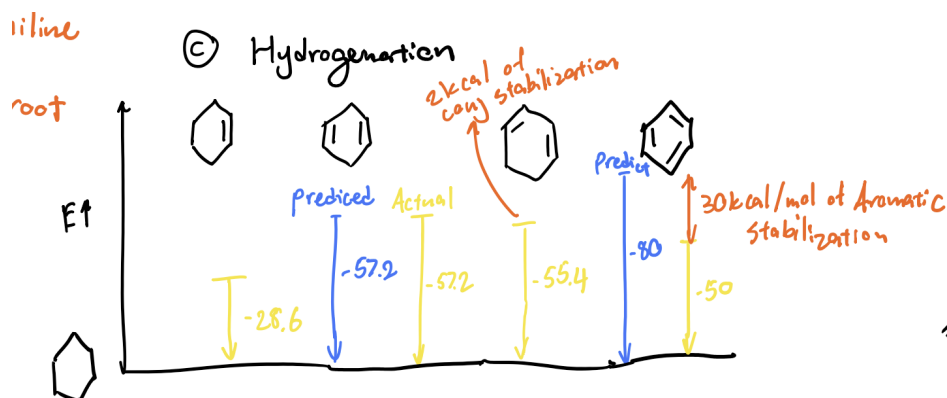


Figure 5

In order to explain this, let's look at a top down view of the benzene molecular orbital diagram.

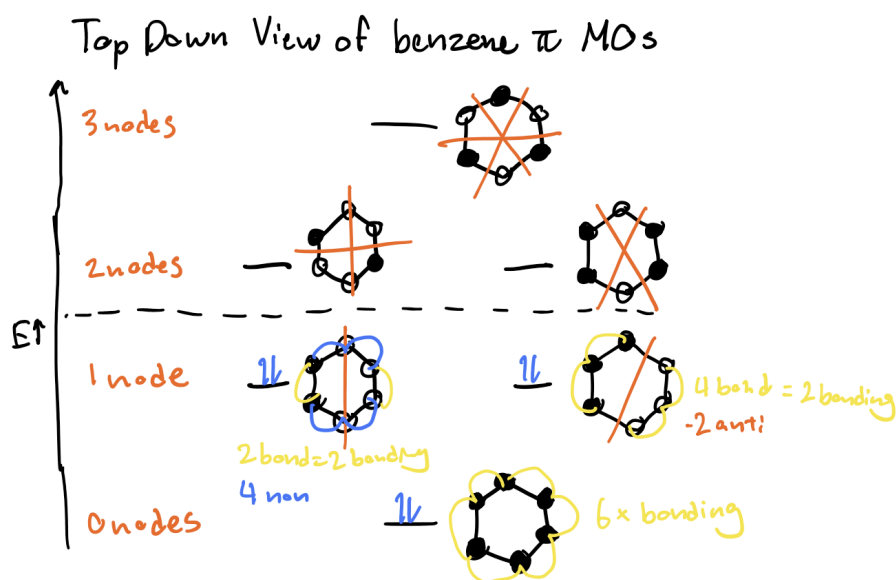


Figure 6

explanation

Core Concept 1.4: Aromatic Compound Molecular Diagrams

The **Frost Diagram** is a method to quickly predict π molecular orbital energy levels for a cyclic conjugated system, following the following steps.

- (1) Compound must have acyclic array of p orbitals
- (2) Draw shape pointed down

- (3) Draw a circle around the shape
- (4) Intersections are energy levels
- (5) Middle of circle is nonbonding
- (6) Add e^- s

Here are the respective orbital diagram examples for a pentene anion and a heptene cation. Benzene has already been shown in FIG REFERENCE ABOVE.

(a) **Pentene Anion**

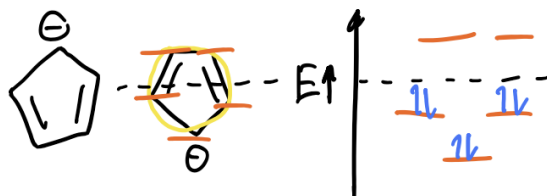


Figure 7

(b) **Heptene Cation**

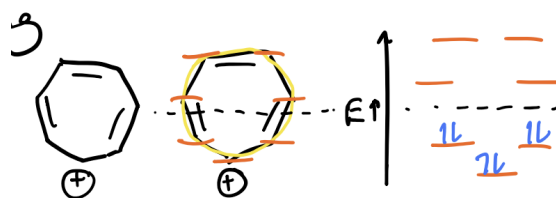


Figure 8

Core Concept 1.5: Aromaticity Rules

In order for a compound to be considered aromatic, it must follow a strict set of rules.

- (1) Cyclic array of p orbitals
- (2) e^- count in cyclic array fulfills the equation of $4n + 2$, where n is an integer (0, 1, 2, 3, 4, ...).

*An e^- count that fulfills the equation of $4n$ where n is an integer is much more reactive than predicted and is termed **anti-aromatic***

If the above criteria are not met, the compound is considered **nonaromatic**, such as the cyclohexene shown below.

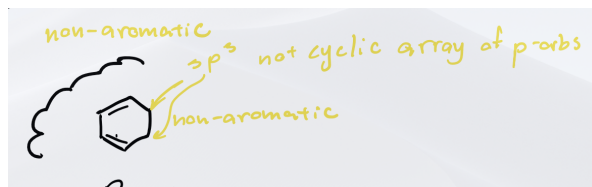


Figure 9

Figure 10: caption

As practice, take a look at the follow two examples.

(a) **Pentene Anion**

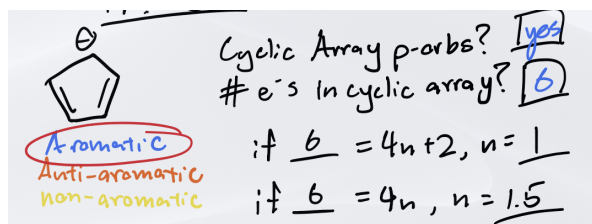


Figure 11

Figure 12: caption

This is a cyclic array of p orbitals and the e^- count is 6, which fulfills the $4n + 2$ expression. Thus this compound is **considered aromatic**.

(b) **Pentene Cation**

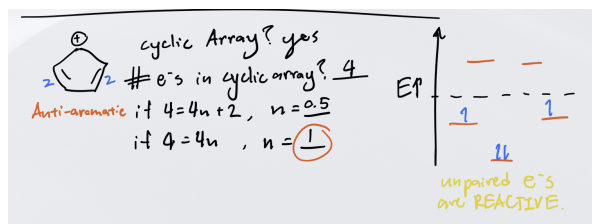


Figure 13

Figure 14: caption

This compound is also a cyclic array of e^- s but its electron count is 4 which does not fulfill the expression $4n + 2$ but rather $4n$. Thus this compound is considered **anti-aromatic**.

Notice that the corresponding molecular orbital diagram results in two unpaired e^- s, making it reactive.

Core Concept 1.6: Aromatic Contributors

An **aromatic contributor** refers to a resonance structure of a molecule that possesses a fully conjugated, cyclic, planar π system satisfying the $4n + 2 \pi e^-$ count, even if the dominant or lowest-energy Lewis structure doesn't immediately look aromatic. Take the following molecule for instance.

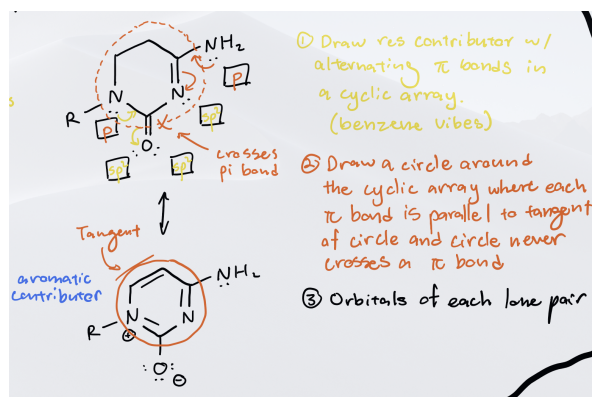


Figure 15: figure of the base non aromatic state to the left and the contributor state to the right

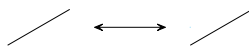


Figure 16: caption

Notice that while the base state is nonaromatic by definition because the conjugated π orbital system is not cyclic, an alternative state accessed through resonance is aromatic, making it an aromatic contributor.

1.2 Electrophilic Aromatic Substitution

Core Concept 1.7: Electrophilic Aromatic Substitution

Electrophilic aromatic substitution (EAS) refers to a reaction wherein an aromatic ring acts as a nucleophile. This reaction type follows the following general mechanism.

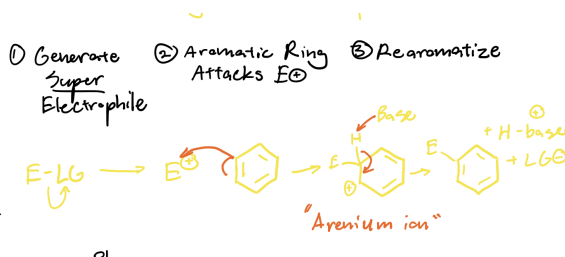


Figure 17

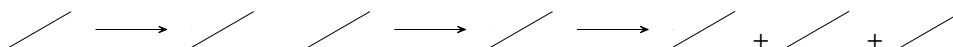


Figure 18: caption

1.2.1 Reaction Types

Core Concept 1.8: Halogenation

EAS halogenation effectively attaches a single halogen to an aromatic ring, generating the corresponding hydrohalogen as a byproduct. The general mechanism utilizes a **Lewis acid** (any chemical species capable of accepting a pair of electrons) as shown below.

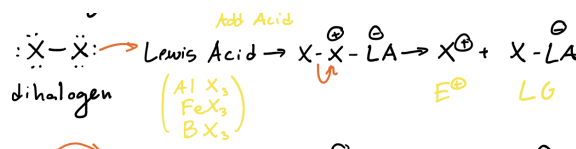


Figure 19

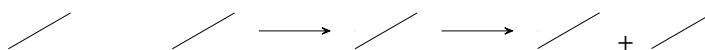


Figure 20: caption

Analogously, here are examples showcasing chlorination and bromination of benzene.

(a) Chlorination of Benzene

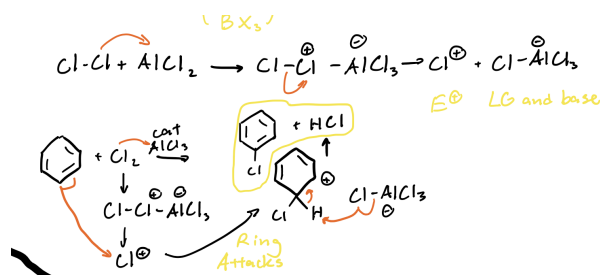


Figure 21

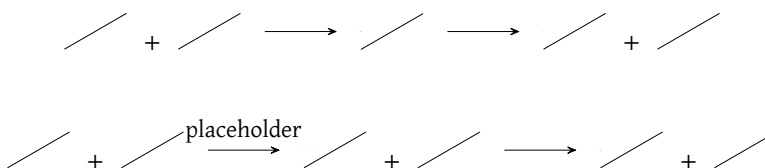


Figure 22: caption

(b) Bromination of Benzene

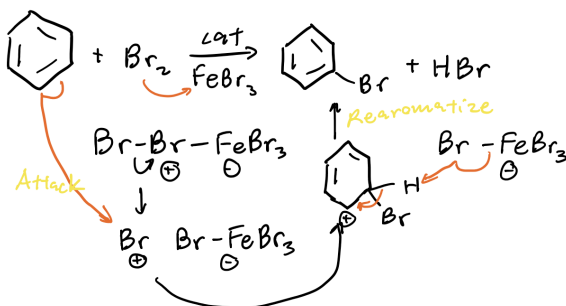


Figure 23

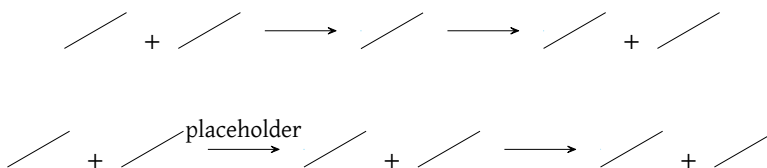


Figure 24: caption

Core Concept 1.9: Nitration

Nitration effectively attaches an NO_2 nitro group onto an aromatic ring, acting primarily as a synthetic stepping stone in practice. First, let's look at the generation of an NO_2^+ electrophile through the exposure of nitric acid to sulfuric acid.

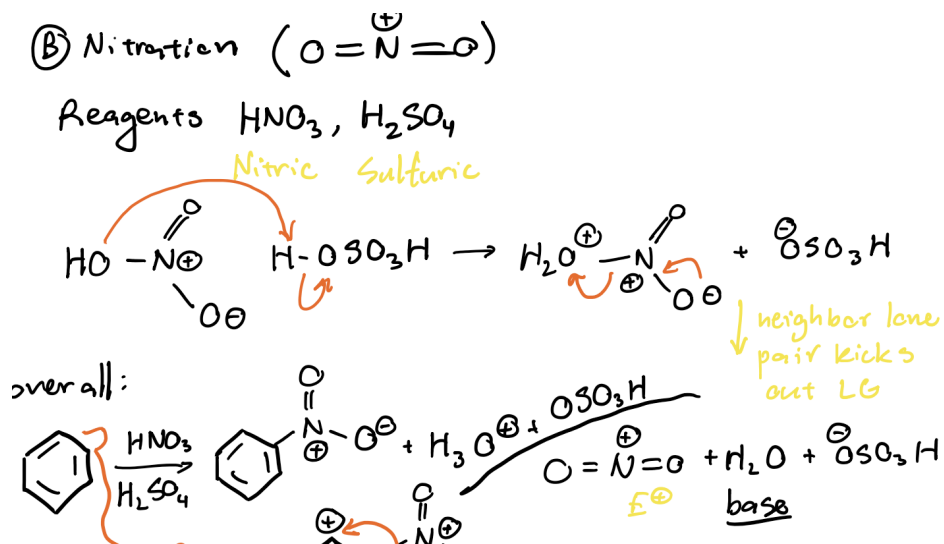


Figure 25



Figure 26: caption

With this in mind, the general nitration mechanism to a benzene follows the outline reaction mapped out in Core Concept 1.7.

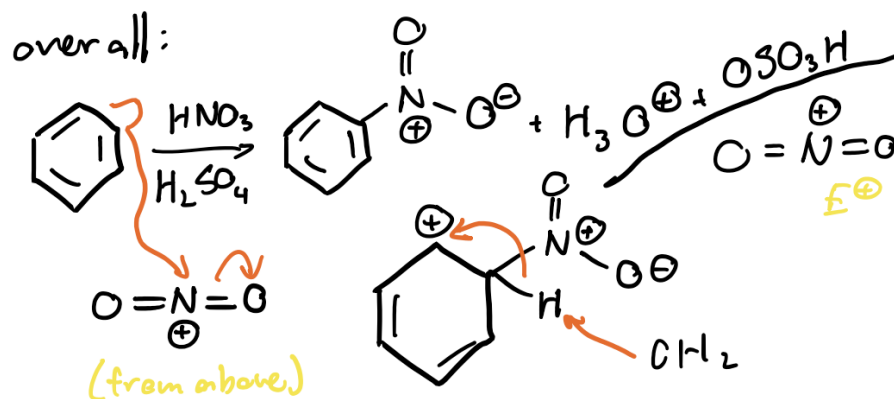


Figure 27



Figure 28: caption

Core Concept 1.10: Sulfonation

Sulfonation entails the attachment of a SO_3H group onto an aromatic ring. The first step is a reaction between two H_2SO_4 molecules, a consequence of one of the reagents being **fumming** H_2SO_4 .

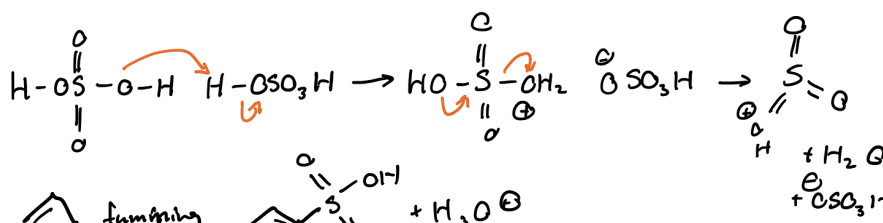


Figure 29



Figure 30: caption

Then follows the same general EAS reaction mechanism covered in Core Concept 1.7.

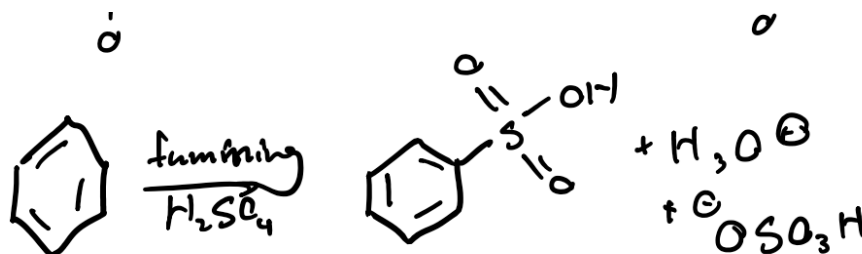


Figure 31



Figure 32: caption

Core Concept 1.11: Alkylation

Alkylation is the process of adding an alkyl group onto an aromatic structure. In order for an alkane to become electrophilic, it must undergo a preparatory conversion to a carbocation state through the following reaction.

(a) placeholder

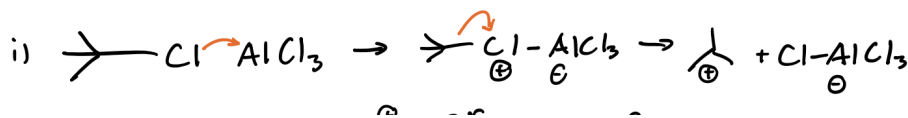


Figure 33



Figure 34: caption

This reaction is similar to the initial reaction shown in the chlorination pathway within Core Concept 1.8.

(b) placeholder

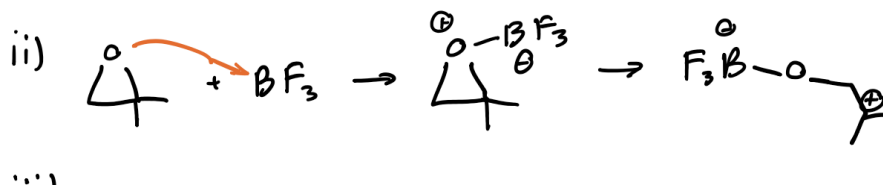


Figure 35

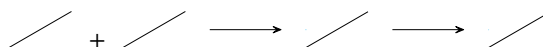


Figure 36: caption

(c) placeholder

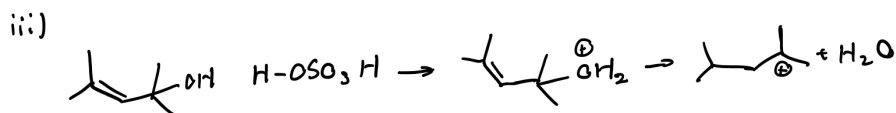


Figure 37



Figure 38: caption

(d) placeholder

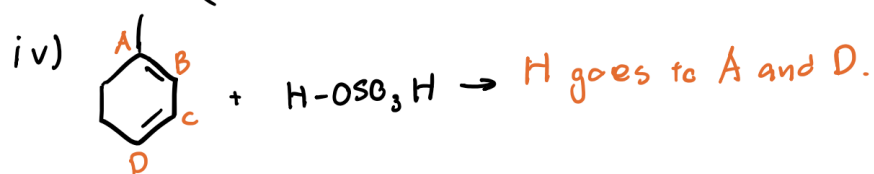


Figure 39



Figure 40: caption

In certain instances, carbocation rearrangements can occur when the directly resulting carbocation is unstable.

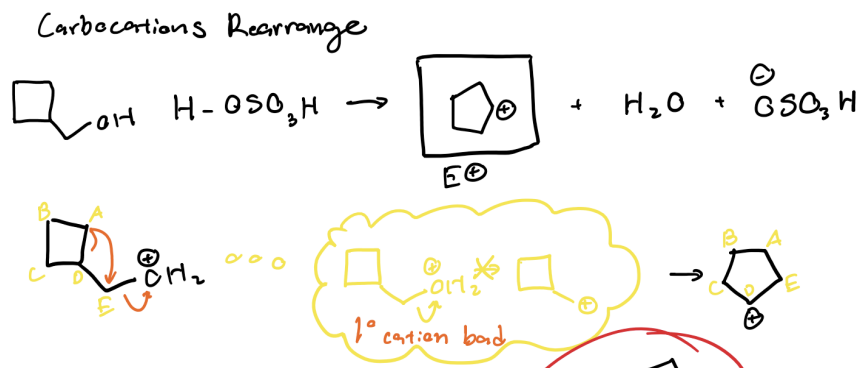


Figure 41

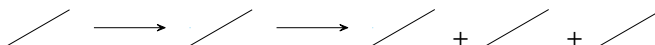


Figure 42: caption

Subsequently, these carbocation alkanes then proceed to perform EAS to aromatic structures.

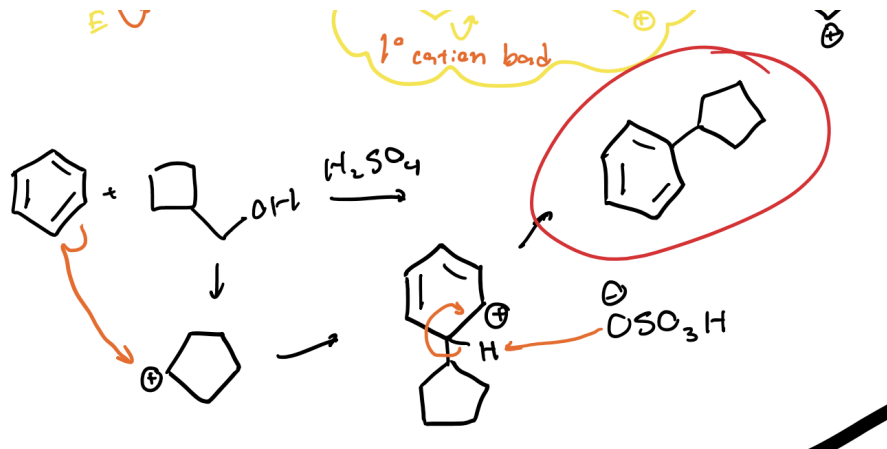


Figure 43



Figure 44: caption

In the specific case above showcasing carbocation rearrangement of a cyclobutane to a cyclopentane, we can prevent rearrangement via a stepwise reaction pathway of **acylation** (Core Concept 1.12) and then reduction with Zn(Hg) and HCl.

(a) Acylation

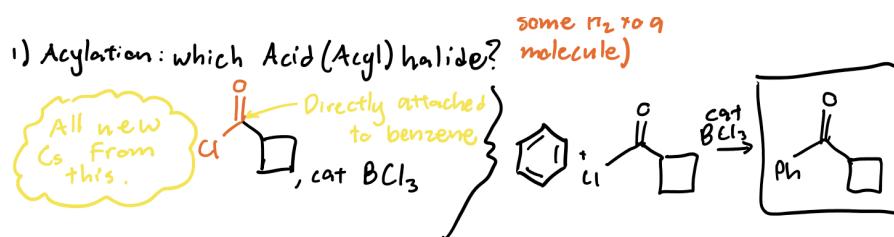


Figure 45



Figure 46: caption

(b) Reduction

2) Reduction * memorize Reagents *

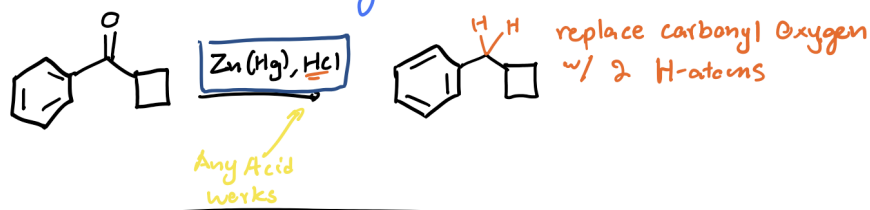


Figure 47

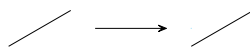


Figure 48: caption

Core Concept 1.12: Acylation

Acylation is the process of adding an acyl halide compound to an aromatic structure. We first need to make the acyl halide electrophilic through a lewis acid as shown in the following general reaction outline.

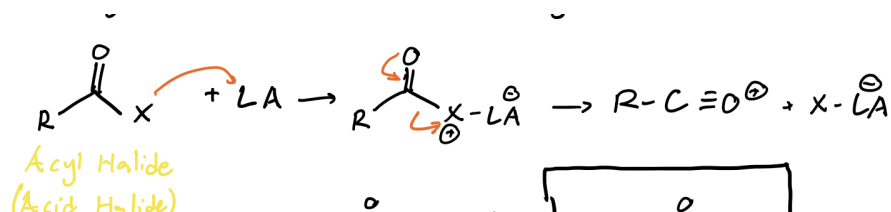
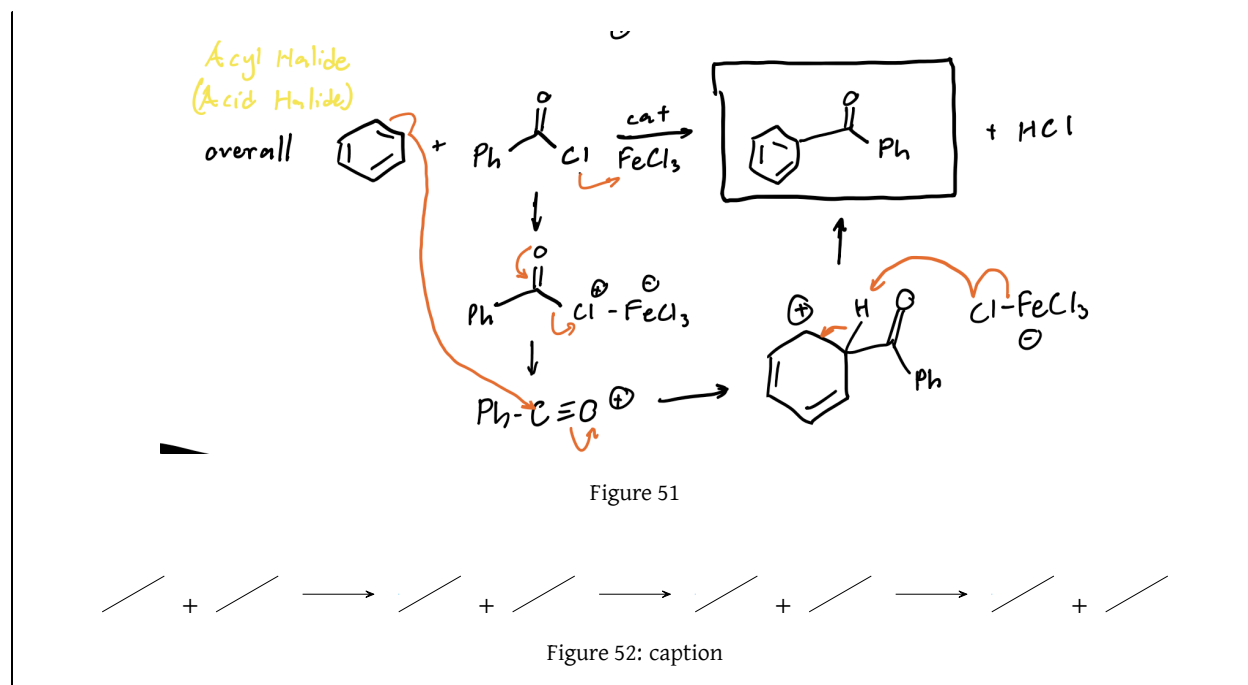


Figure 49



Figure 50: caption

The resulting RCO^+ molecule can then perform EAS with an aromatic ring. A full reaction using real compounds is shown below.



1.2.2 Directing Groups

Core Concept 1.13: Directing Groups

So far, we have covered simply the EAS reaction types in Subsubsection 1.2.1 namely using benzene as our sole relevant aromatic structure. However, in scenarios where the aromatic ring is asymmetrical, chemists need a method to determine which carbon they want the EAS to occur on. Now we introduce **directing groups**, functional groups already attached to a benzene ring that dictate where (O, M, P) a new incoming electrophile will land.

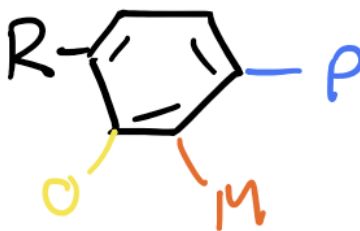


Figure 53

Figure 54: caption

A summary table is provided below, but visit each Core Concept for a more detailed look into each directing group type.

Directing Group Type	Examples	Description
(1) Inductive O/P Director	$R = \text{alkyl [Me, Et, } i\text{Pr, ...]}$	Alkyl groups donate electron density inductively, directing substitution to the ortho and para positions.
(2) Inductive M Director	$R = \text{carbon w/ EN atoms [CCl}_3, \text{CF}_3, \text{...}]}$	Carbons bearing electronegative atoms withdraw electron density inductively, directing substitution to the meta position.
(3) Resonance O/P Director	i) R has lone pair adjacent to ring: [OMe, SR, NMe ₂] ii) $R = \text{carbon w/ } \pi \text{ system: [alkene, alkyne, benzene]}$	Substituents that donate electron density via resonance into the ring direct substitution to the ortho and para positions.
(4) Resonance M Director	[C=O, C≡N, NO ₂]	Substituents that withdraw electron density via resonance from the ring accept electron density and direct substitution to the meta position.

Table 1: Classification of electrophilic aromatic substitution directing groups

Core Concept 1.14: Inductive O/P Directors

Inductive O/P directors are substituents that direct any EAS to the O and P positions within an aromatic ring.

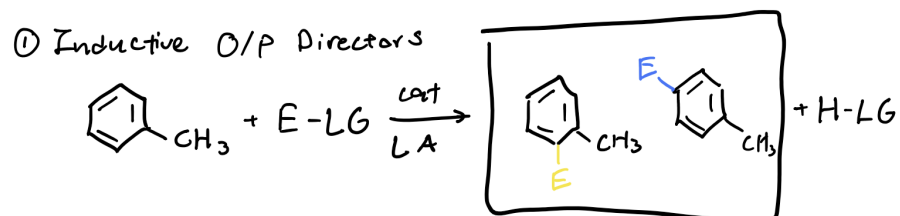


Figure 55

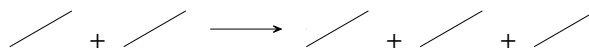


Figure 56: caption

The substituents are generally alkyls, and the mechanism behind why they favor O/P over M positioning is due to the general stability of the different resonance structures that occur in each positioning scenario.

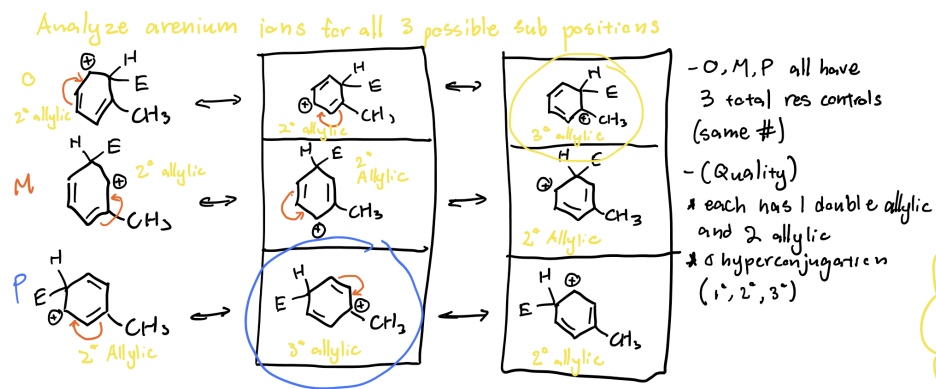


Figure 57

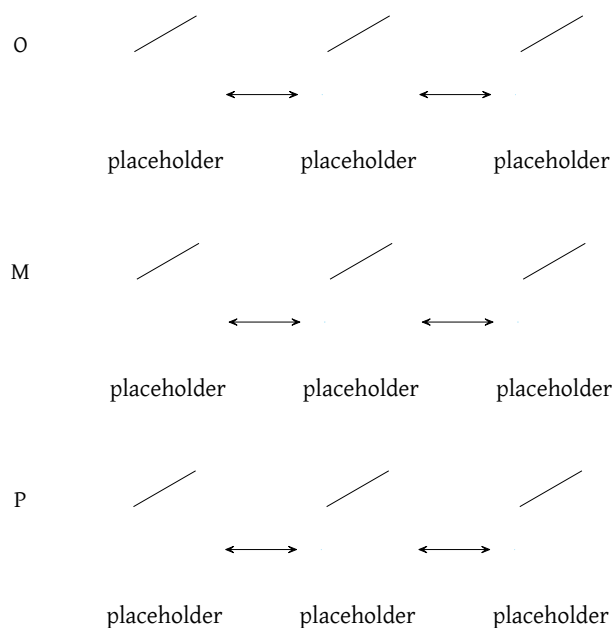


Figure 58: caption

As shown above, while nearly all of the different possible resonance structures place the transient carbocation charge on a 2° allylic position, O/P positioning allows for a 3° allylic carbocation position within 1 out of 3 resonance structures for each whereas M positioning does not. As such the energy levels of the resulting carbocation intermediaries follow the general ordering shown in the energy diagram below.

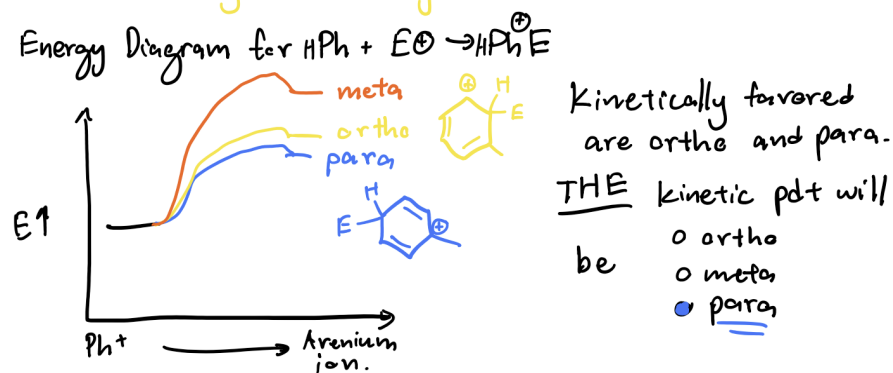


Figure 59

Core Concept 1.15: Inductive M Directors

Inductive M directors are substituents that direct any EAS to the M position within an aromatic ring. The substituents are carbons bearing highly electronegative atoms, making the carbon highly partially positive.

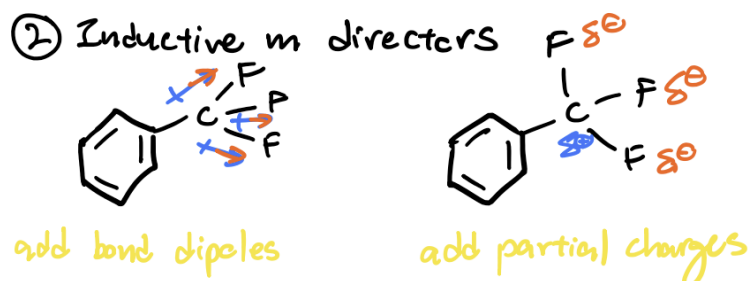


Figure 60

Figure 61: caption

The M position is favored because the O/P 3° allylic carbocation resonance structures have possess extra unstable repulsion between the carbocation and the partial positive carbon within the substituent. The M positioning, on the other hand, possess for a resonance structure that allows for maximal distance between the positive substituent carbon and the carbocation charge on teh opposite side fo the aromatic ring.

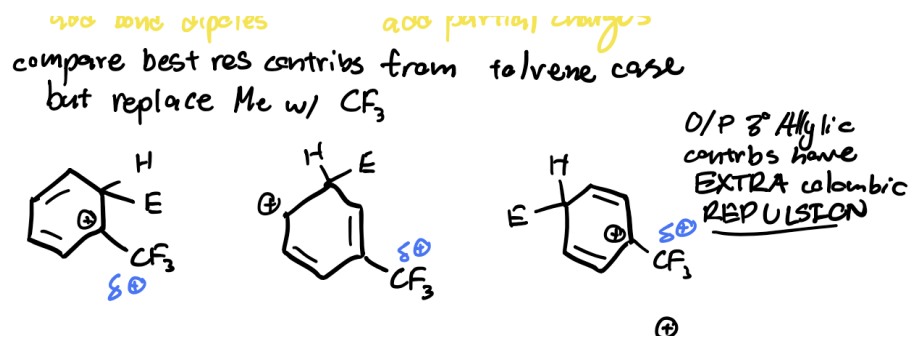


Figure 62

Figure 63: caption

Consequently, the resulting energy levels put O/P positioning energy above that of M positioning.

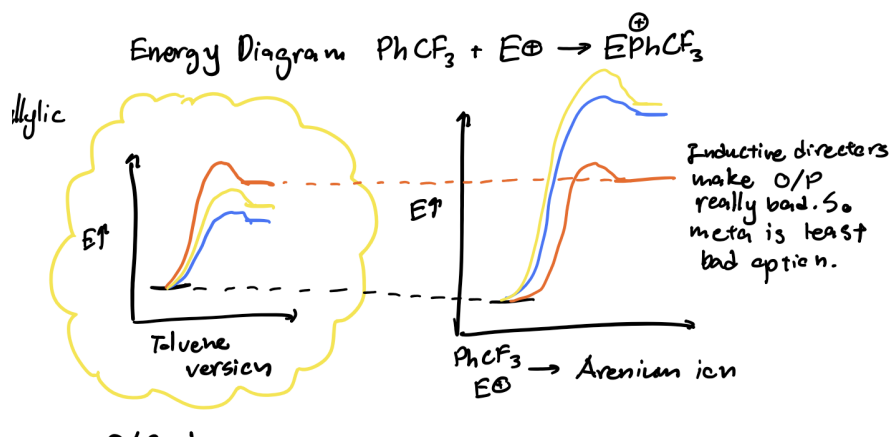


Figure 64

Core Concept 1.16: Resonance O/P Directors

Resonance O/P directors are substituents that direct any EAS to the O and P positions within an aromatic ring through donating e^- density. For instance, phenol possess resonance structures that place a carboanion negative charge on exclusively the O and P positions.

③ Study the NUCLEOPHILICITY
of aromatic Ring.

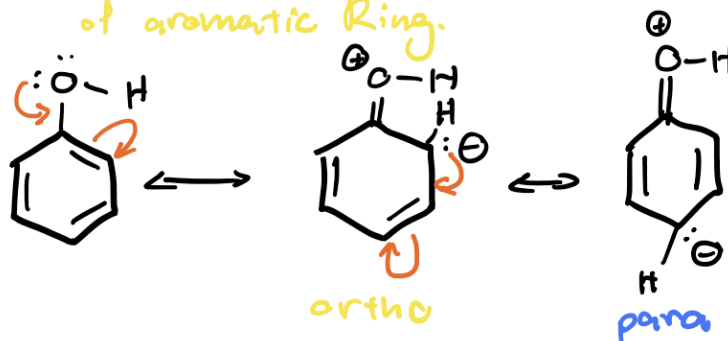


Figure 65

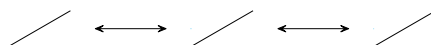


Figure 66: caption

These types of substituents specifically need to be lone pair adjacent to the aromatic ring. Another substituent type that allows for e^- donation is carbon with conjugated π systems. Take the following alkene for instance.

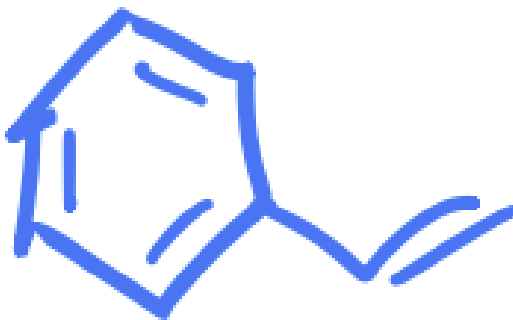


Figure 67: show the resonance possible



Figure 68: caption

Notice that the primary carbocation is able to form because it is allylic.

Here is a side note regarding the method by which inductive directors induce favorability of EAS positioning in comparison to resonance directors -> resonance directors affect the nucleophilicity capabilities of the relevant positions whereas the inductive versions manipulate favorability through transition state stability.

Core Concept 1.17: Resonance M Directors

Resonance M directors are substituents that direct any EAS to the M position within an aromatic ring through withdrawing e^- density. For instance, carbonyl substituents have the ability to withdraw e^- density in a manner that results in subsequent resonance structures to place a carbocation positive charge on the O/P positions, making these locations effectively less nucleophilic and the P position relatively more nucleophilic.

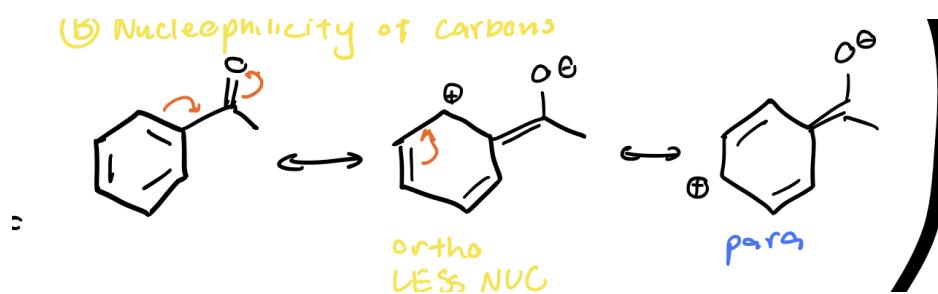


Figure 69

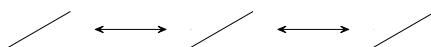


Figure 70: caption

As a result, when an electrophile approaches the transient aromatic ring, the nucleophilic attack occurs most favorably from the M position rather than the O/P position.

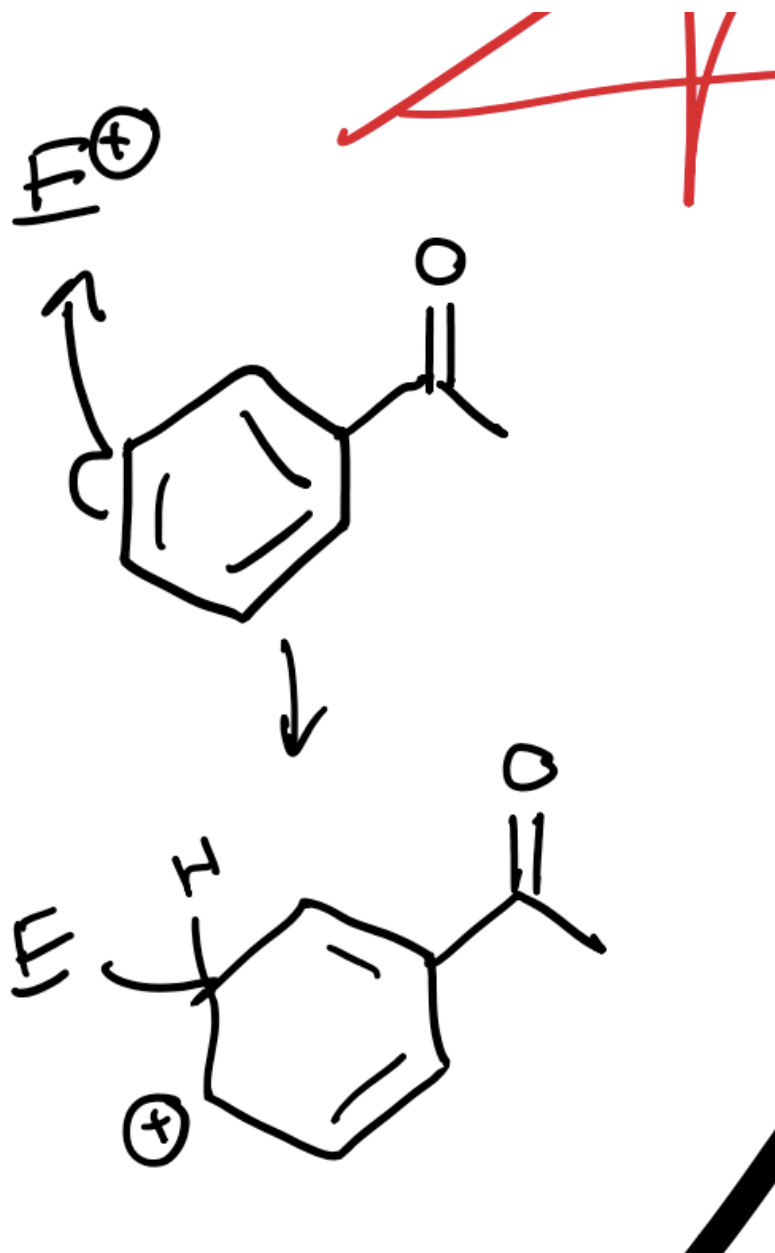


Figure 71



Figure 72: caption

Core Concept 1.18: Halogens as Directing Groups

Halogens can also act as directing groups, and based on data, they act as O/P directors. To understand the underlying basis of this observation, let's analyze monohalobenzene in the context of the inductive effect and

resonance.

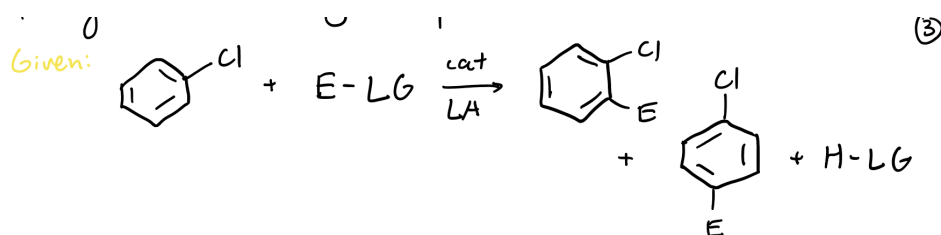


Figure 73

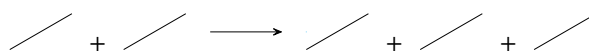


Figure 74: caption

(a) Inductive Effect

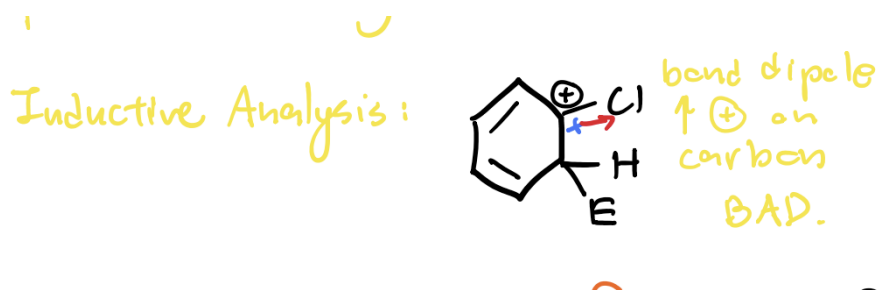


Figure 75



Figure 76: caption

Under an O/P directing EAS interaction, inductive analysis would actually discourage O/P directing because the subsequent carbocation transition state places a carbocation that already has lowered e^- density even lower by residing next to the highly electronegative halogen atom. As a result, the dipole moment created destabilizes the structure.

(b) Resonance

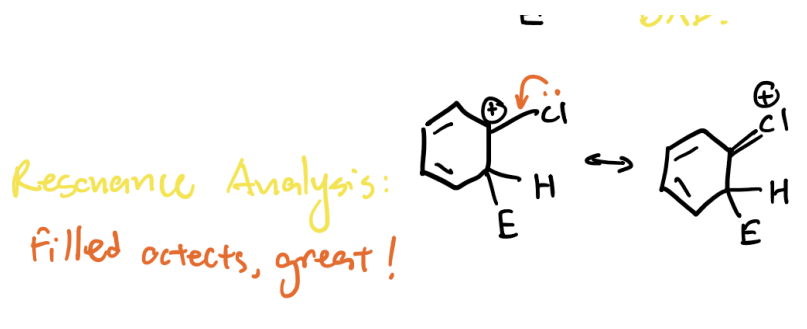


Figure 77

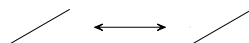


Figure 78: caption

Resonance, on the other hand, allows the halogen to donate e^- density in order to fill all octet requirements throughout the entire molecule, which favors the O/P directing EAS interaction.

Although these two effects contradict one another, we can confidently say that the effects of resonance trump the inductive effect because real data in practice results in the O/P directing as favored.

1.2.3 Aromatic Substituent Manipulations

Core Concept 1.19: Reduction of Acyl Ketone

The reduction of an acyl ketone is a reduction reaction that was previously used in Core Concept 1.11 to replace a carbonyl oxygen with two hydrogen bonds.

① Reduction of Acyl ketone

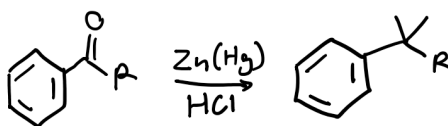


Figure 79



Figure 80: caption

No mechanism is provided for this reaction.

Core Concept 1.20: Nitrobenzene Reduction to Aniline

The reduction of nitrobenzene work similarly to the reduction of acyl ketone (Core Concept 1.19) wherein oxygens are replaced by hydrogens. This reaction uses the same reactants to carry this replacement out.

② Nitro Reduction

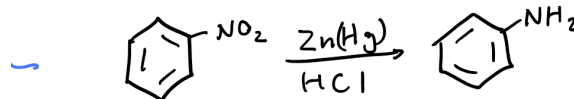


Figure 81



Figure 82: caption

No mechanism is provided for this reaction.

Core Concept 1.21: Diazotization of Aniline

The diazotization of aniline is a potential subsequent reaction to nitrobenzene reduction as the product of the latter is the reactant of the former.

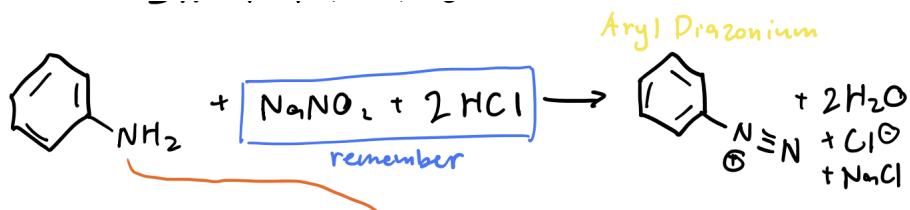


Figure 83

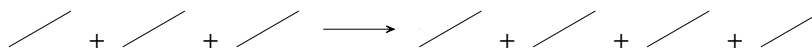


Figure 84: caption

The reaction mechanism occurs as follows, first starting with the conversion of NO_2^- to a nitrosyl cation.

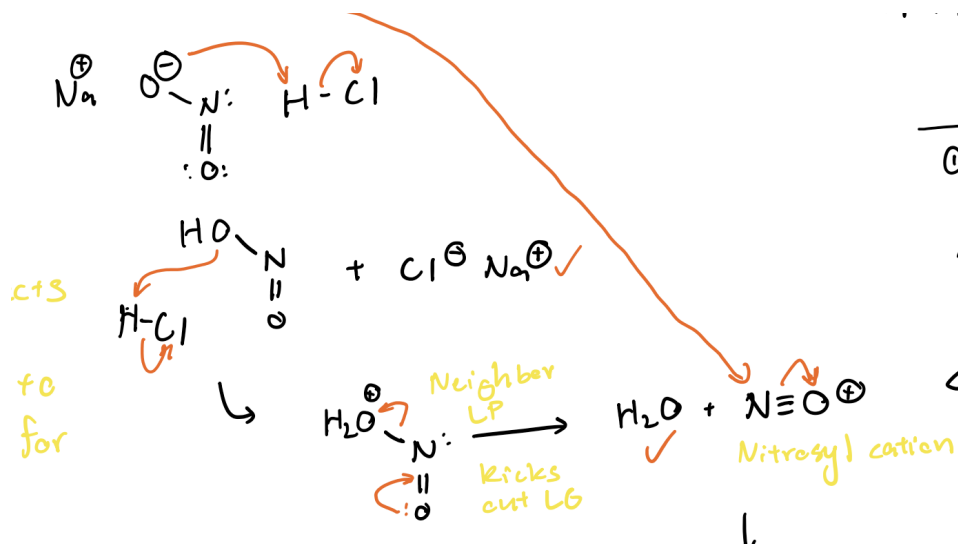


Figure 85

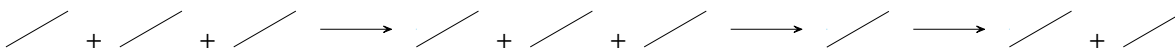
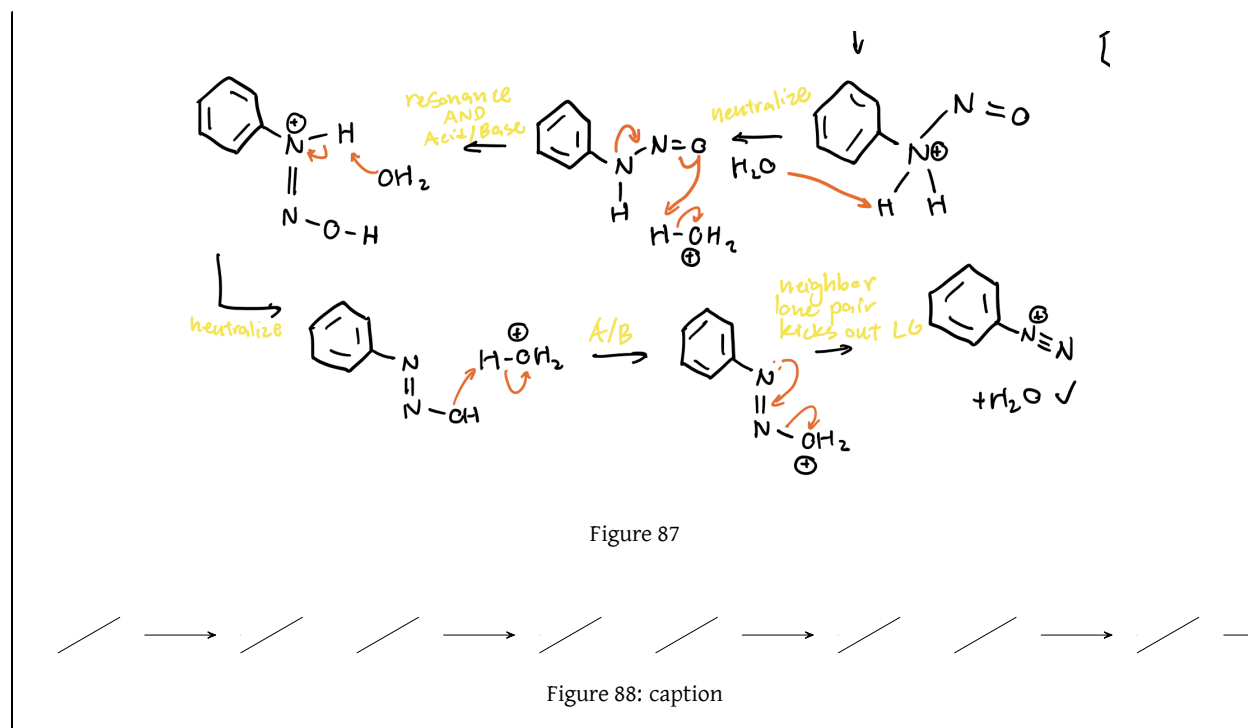


Figure 86: caption

The highly reactive nitrosyl cation then goes on to interact with the aniline in the following manner to produce **aryl diazonium**.



1.3 Nucleophilic Aromatic Substitution

Core Concept 1.22: Nucleophilic Aromatic Substitution

Nucleophilic aromatic substitution (NAS or $\text{S}_{\text{N}}\text{Ar}$) is an interaction in which the aromatic ring is the electrophile. NAS comes in two "flavors": **diazonium** (Core Concept 1.23) or **addition/elimination** (Core Concept 1.24) chemistry.

Recall that in electrophilic aromatic substitution (Subsection 1.2), the aromatic ring is the nucleophile.

Core Concept 1.23: Diazonium Chemistry

Diazonium chemistry begins by N_2 leaving a phenyldiazonium molecule—via breaking the bond between the σ $\text{C}(\text{sp}^2)\text{--N}(\text{sp})$ bond—to produce a carbocation possessing benzene ring and an extremely stable N_2 gas molecule.

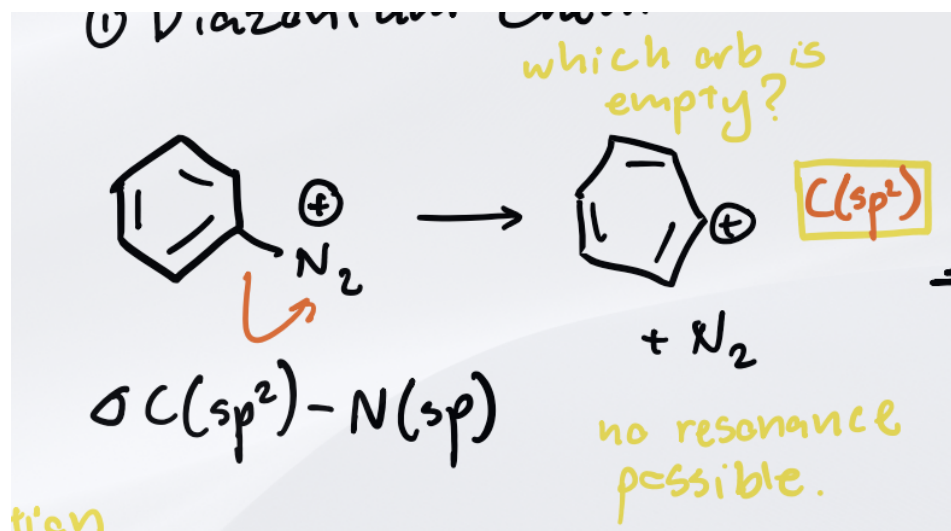


Figure 89

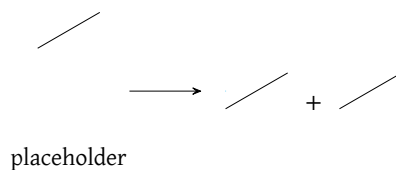


Figure 90: caption

It's important to notice that resonance is not possible in this specific carbocation state because the orbital that is empty is the $\text{C}(\text{sp}^2)$ orbital. We would need an unbalanced p orbital for resonance to occur. As a result, this phenyl cation is highly reactive and hungry for any nearby nucleophile to help stabilize its positive charge.

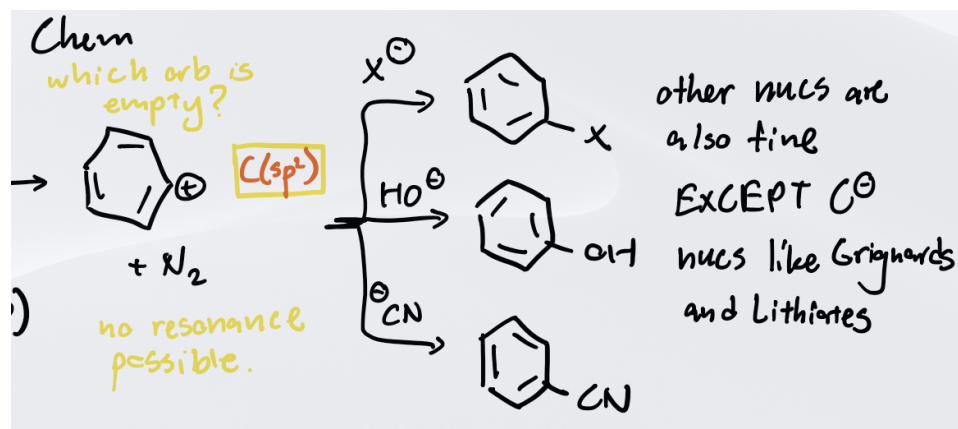


Figure 91

figure 92: caption

Figure 92: caption

As shown above, nearly all covered nucleophiles work in this interaction **with the exception of grignards and lithiates**. The reasoning behind this will not be covered in the scope of these notes.

Combining this diazonium chemistry with previously learned reactions, chemists can effectively attach nearly any nucleophile to a benzene molecule using the following reaction pathway.

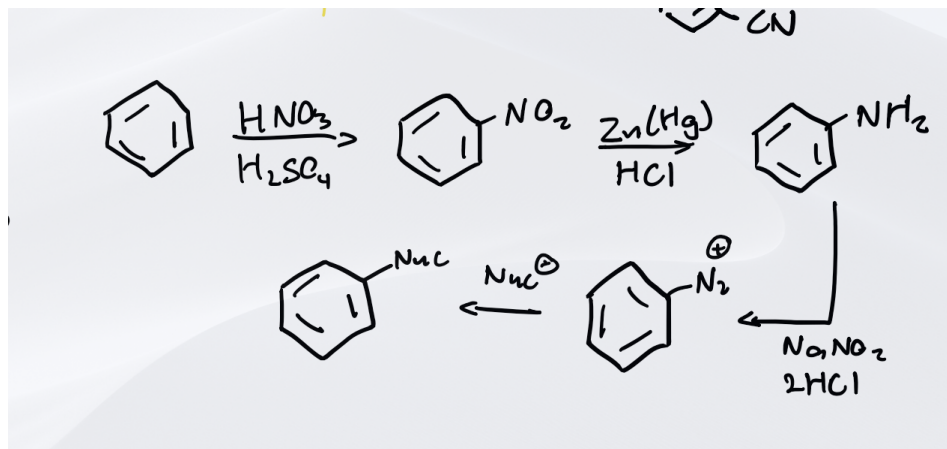


Figure 93: reference the core concepts for the reactions within the caption



Figure 94: caption

Core Concept 1.24: Addition/Elimination

The NAS **addition/elimination** reaction occurs in the following general manner.

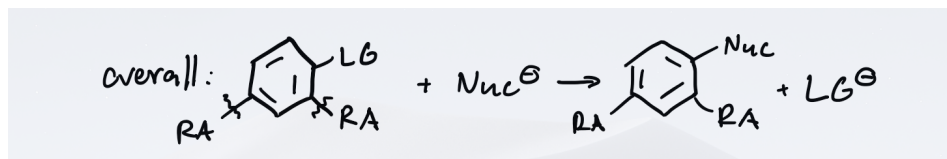


Figure 95



Figure 96: caption

Recall that we have seen resonance acceptors in resonance M directors (*Core Concept 1.17*), however, these resonance acceptors are not for the purpose of directing meta in the context of NAS chemistry. Their main purpose is to be able to accept e^- density after the nucleophile attacks the carbon with the leaving group.

Additionally, this reaction requires strong nucleophiles because it ultimately needs to break aromaticity as shown in the mechanism below. It should be noted that this step—the addition—is consequently the rate-determining step.

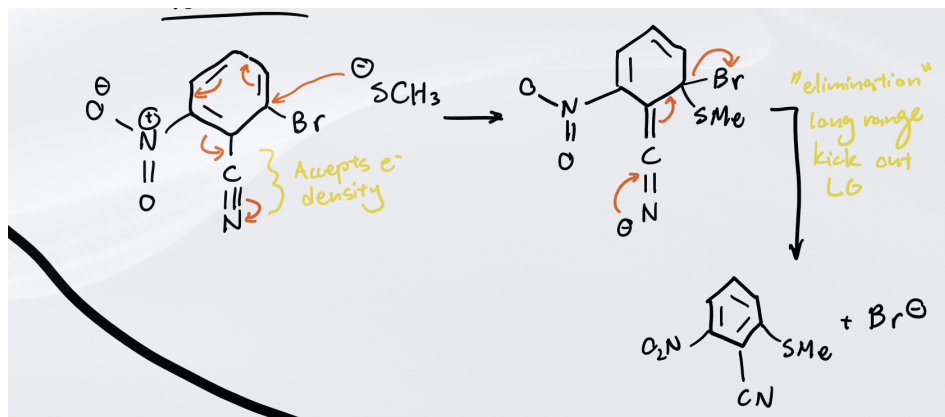


Figure 97



Figure 98: caption

As the mechanism showcases, the reaction is simply a relay of e^- density to the resonance acceptor, which then bounces back the e^- density in order to kick out the leaving group in the end. However, since the outcome of this interaction is ultimately the same as that of an S_N2 reaction, one may wonder why S_N2 is not a viable pathway.

Revisiting the S_N2 mechanism, recall that, in the lense of molecular orbitals, the nucleophile interacts with the σ^* anti-bonding orbital between the electrophile and leaving group in order to kick out the leaving group and create a bond simultaneously via a backside attack.

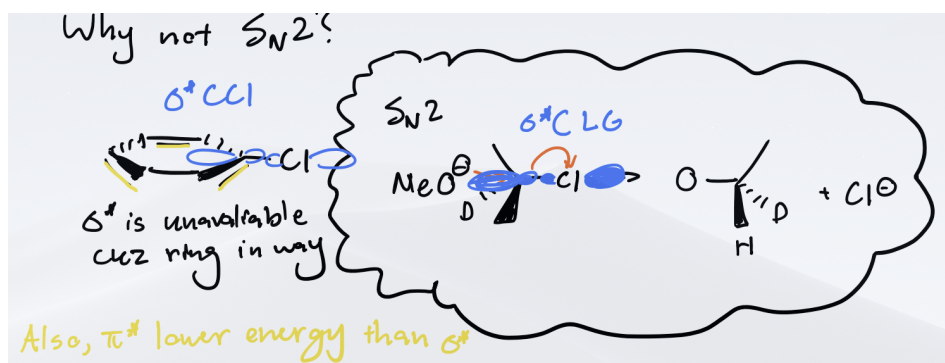


Figure 99

Figure 100: caption

In the aromatic ring scenario, this necessary σ^* interaction is extremely unlikely due to the physical barrier to entry because of the ring itself.

In the context of the frontier molecular orbital theory, the π^* orbital is lower in energy than σ^* , making it the LUMO and a much more accessible orbital to interact with, which is what happens in addition/elimination, the favorable pathway.

Lets go through two examples.

(a) **Example 1: Predicting Major Kinetic Product**

Consider the following reaction consisting to two possible leaving groups F and Cl with a NO_2 resonance acceptor.

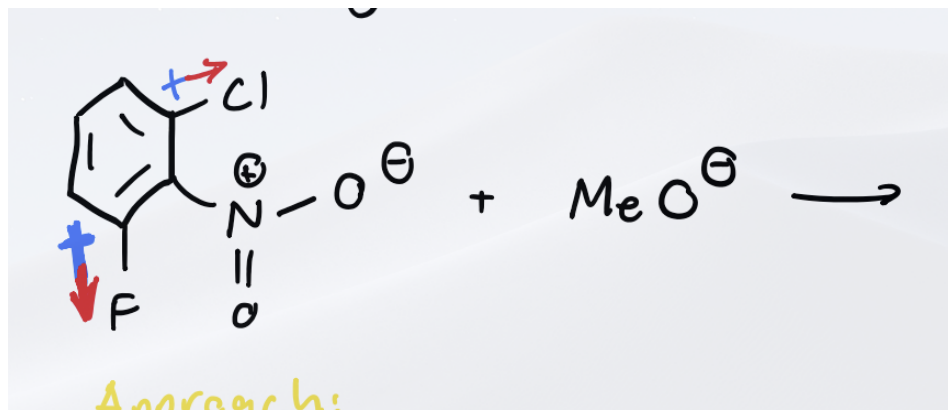


Figure 101



Figure 102: caption

Both addition/elimination pathways are possible as shown below, ultimately leading to two distinct products.

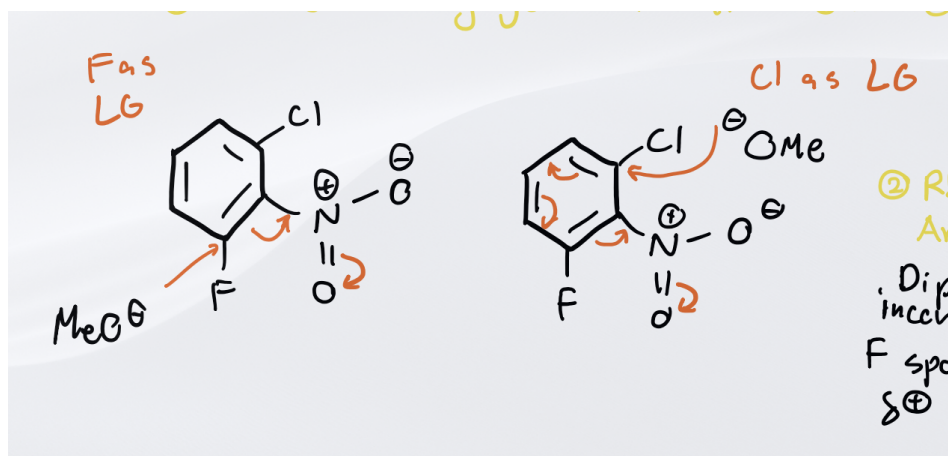


Figure 103



Figure 104: caption

Since the nucleophile is held constant across both reaction pathways, the ability of the nucleophile to break aromaticity is not a factor. Instead, it is the greater dipole moment created by the more electronegative

fluorine over the chlorine. Due to the greater partial positive charge of the fluorine carbon, MeO^- favors attacking that carbon because it is more electrophilic. Thus the reaction favors the substitution of the F over the Cl.

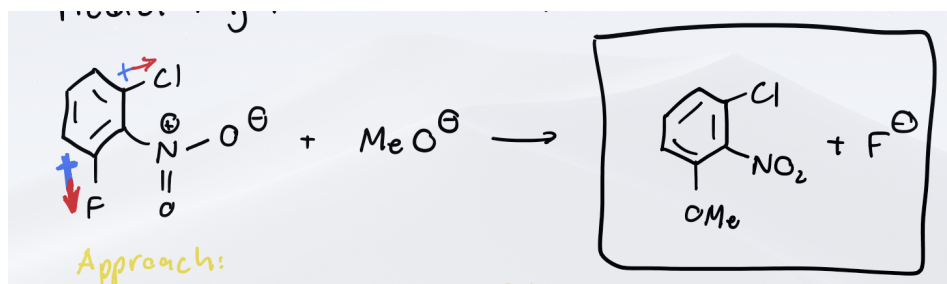


Figure 105



Figure 106: caption

(b) **Example 2: Predicting Product of NAS Chemistry**

Consider the following addition/elimination reaction below.

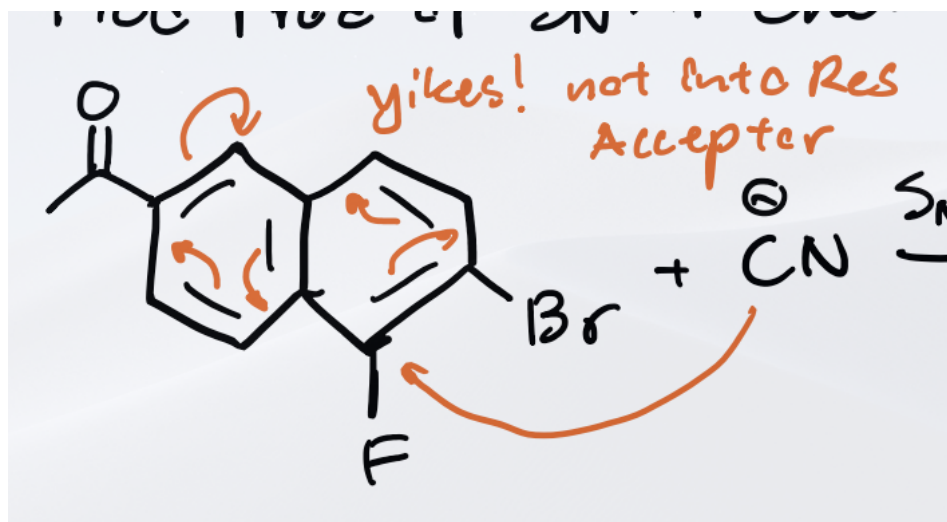


Figure 107: short initial reactants without arrows



Figure 108: caption

In this case, simply only one of the leaving group substitutions work to donate e^- density into the resonance acceptor and the other doesn't.

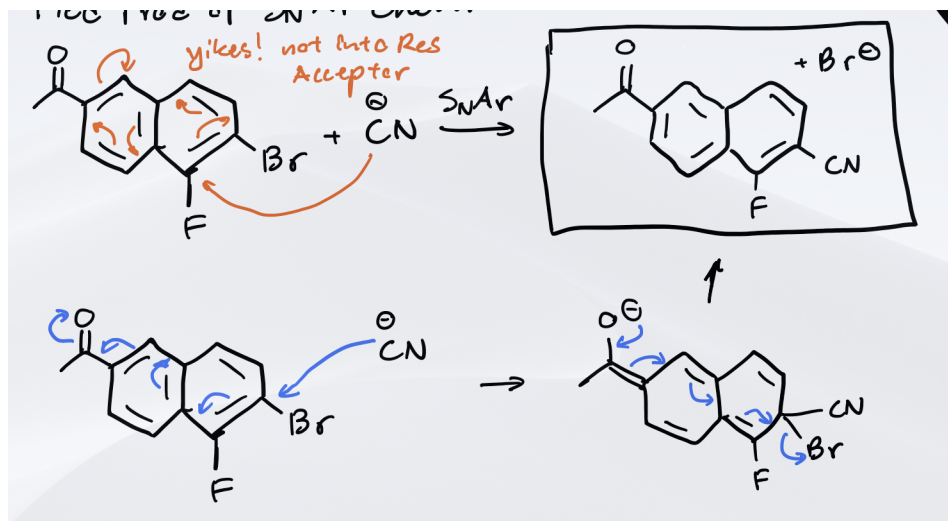


Figure 109: showcase the pathways and how one isn't possible

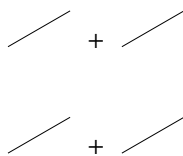


Figure 110: caption

Thus the exclusive reaction outcome maps out as follows.

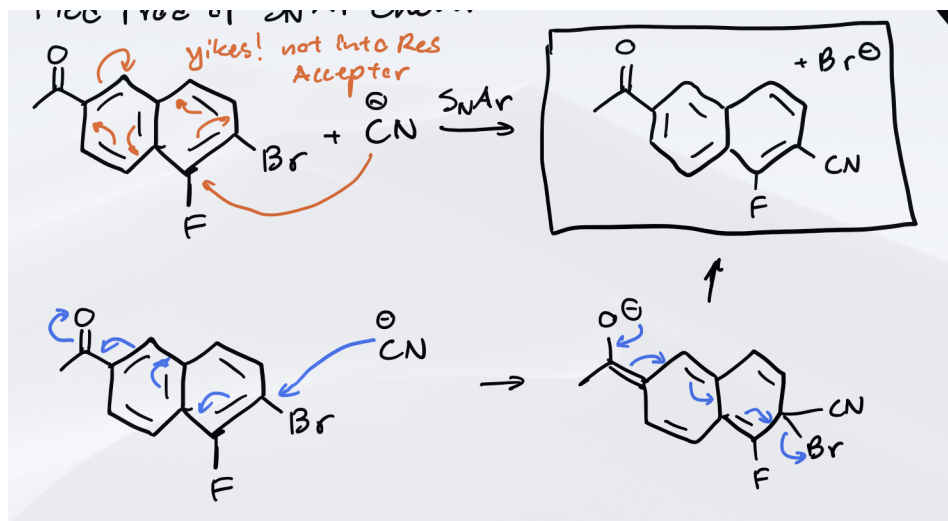


Figure 111: complete chemical equation

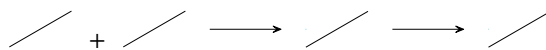


Figure 112: caption

(c) **Example 3: Predicting Starting Material**

Consider having to determine the starting aromatic compound to the following products based on the given nucleophile.

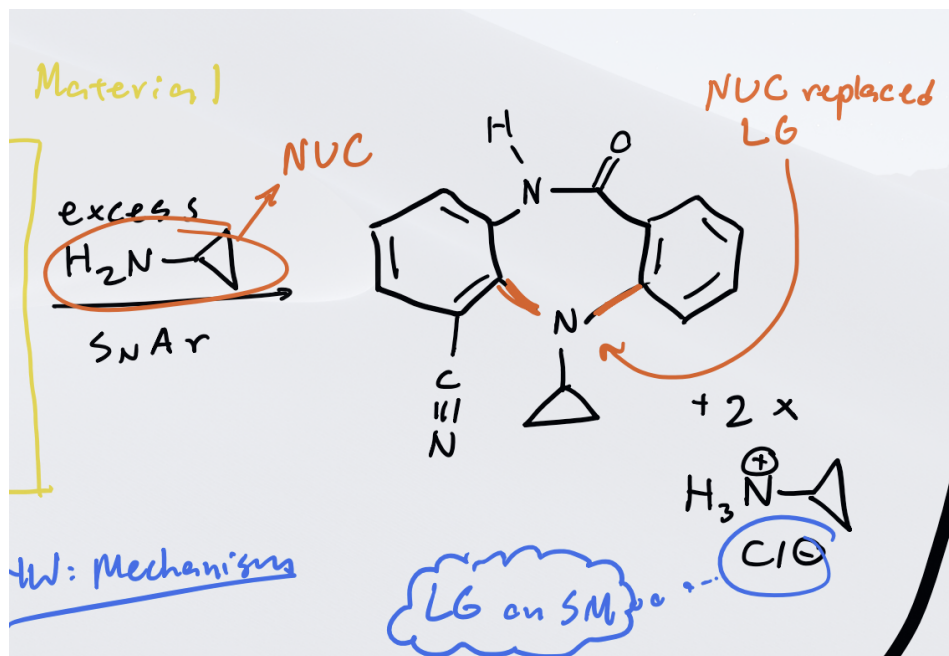


Figure 113

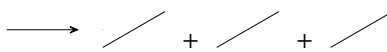


Figure 114: caption

Because we see two Cl^- and two $\text{H}_3\text{N}^+\text{R}$ on the product side, we know that the addition/elimination has occurred twice. Luckily the product quite literally has the nucleophile placed directly on center connected to the aromatic ring system via two bonds on the nitrogen. Based on this, we can reasonably predict the starting material to have had these two bond substituted.

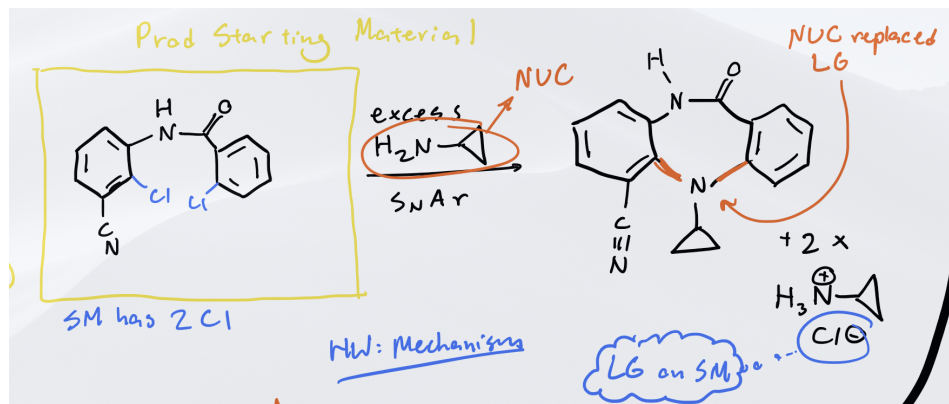


Figure 115



Figure 116: caption